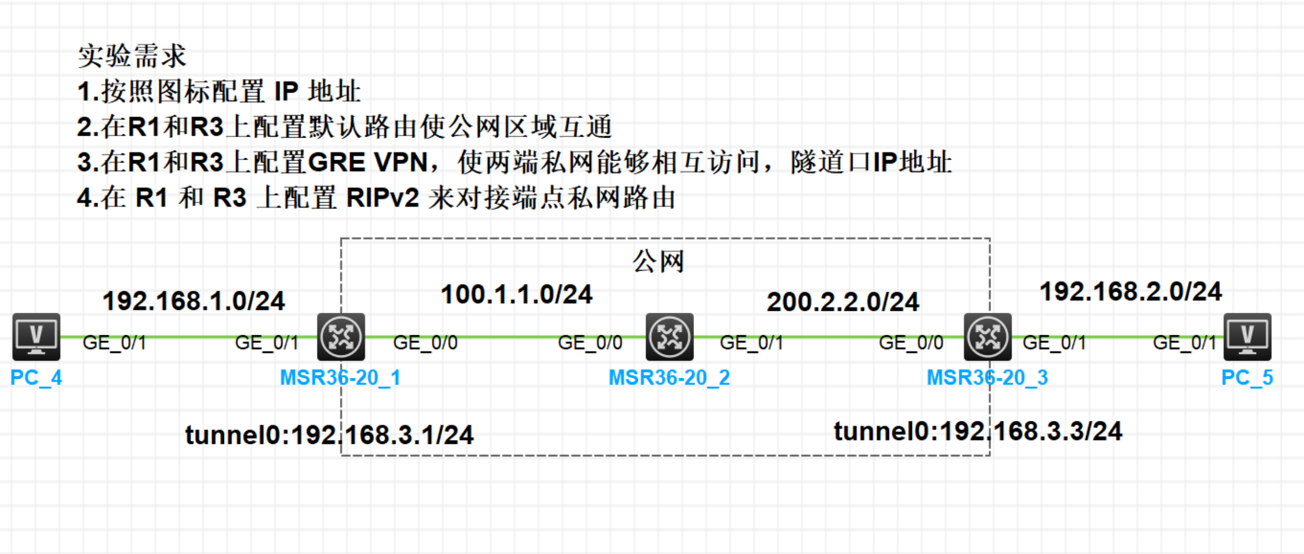
GRE VPN实验

实验拓扑



实验需求

> 1. 按照图标配置 IP 地址
> 2. 在R1和R3上配置默认路由使公网区域互通
> 3. 在R1和R3上配置GRE VPN，使两端私网能够相互访问，隧道口IP地址
> 4. 在 R1 和 R3 上配置 RIPv2 来对接端点私网路由

实验配置

```
[R1]int g0/0
[R1-GigabitEthernet0/0]ip ad 100.1.1.1 24
[R1-GigabitEthernet0/0]int g0/1
[R1-GigabitEthernet0/1]ip ad 192.168.1.1 24
[R1]ip route-static 0.0.0.0 0 100.1.1.2

---
[R2]int g0/0
[R2-GigabitEthernet0/0]ip ad 100.1.1.2 24
[R2-GigabitEthernet0/0]int g0/1
[R2-GigabitEthernet0/1]ip ad 200.2.2.2 24
[R2]ip route-static 192.168.1.0 24 100.1.1.1
[R2]ip route-static 192.168.2.0 24 200.2.2.3
---
[R3]int g0/0
[R3-GigabitEthernet0/0]ip ad 100.2.2.3 24
[R3-GigabitEthernet0/0]int g0/1

[R3-GigabitEthernet0/1]ip ad 192.168.2.1 24
```

```
[R3]ip route-static 0.0.0.0  0 200.2.2.2
######################################################################
[R1]interface Tunnel 0 mode gre
[R1-Tunnel0]ip ad 192.168.3.1 24
[R1-Tunnel0]source GigabitEthernet 0/0
[R1-Tunnel0]destination 200.2.2.3

---
[R3]interface Tunnel 0 mode gre
[R3-Tunnel0]ip ad 192.168.3.3 24
[R3-Tunnel0]source GigabitEthernet 0/0
[R3-Tunnel0]destination 100.1.1.1

############################################################
[R1-Tunnel0]ospf 1
[R1-ospf-1]a 0
[R1-ospf-1-area-0.0.0.0]network  192.168.1.0 0.0.0.255
[R1-ospf-1-area-0.0.0.0]network  192.168.3.0 0.0.0.255

---
[R3-Tunnel0]ospf
[R3-ospf-1] a 0
[R3-ospf-1-area-0.0.0.0]network  192.168.2.0 0.0.0.255
[R3-ospf-1-area-0.0.0.0]network  192.168.3.0 0.0.0.255
```

结果验证

连通性测试

```
<H3C>ping 192.168.2.10
Ping 192.168.2.10 (192.168.2.10): 56 data bytes, press CTRL_C to break
56 bytes from 192.168.2.10: icmp_seq=0 ttl=253 time=2.529 ms
56 bytes from 192.168.2.10: icmp_seq=1 ttl=253 time=1.628 ms
---
<H3C>ping 192.168.1.10
Ping 192.168.1.10 (192.168.1.10): 56 data bytes, press CTRL_C to break
56 bytes from 192.168.1.10: icmp_seq=0 ttl=253 time=1.603 ms
56 bytes from 192.168.1.10: icmp_seq=1 ttl=253 time=1.797 ms
```

查看接口IP

```
<R1>display  ip in brief
*down: administratively down
(s): spoofing  (l): loopback

Interface            Physical Protocol IP address/Mask    VPN instance Description
```

```
GE0/0              up       up       100.1.1.1/24      --            --
GE0/1              up       up       192.168.1.1/24    --            --
Tun0               up       up       192.168.3.1/24    --            --
---
<R2>display ip in brief
*down: administratively down
(s): spoofing  (l): loopback
Interface          Physical Protocol IP address/Mask    VPN instance Description
GE0/0              up       up       100.1.1.2/24      --            --
GE0/1              up       up       200.2.2.2/24      --            --
---
<R3>display  ip in brief
*down: administratively down
(s): spoofing  (l): loopback
Interface          Physical Protocol IP address/Mask    VPN instance Description
GE0/0              up       up       200.2.2.3/24      --            --
GE0/1              up       up       192.168.2.1/24    --            --
Tun0               up       up       192.168.3.3/24    --            --
```

查看OSPF邻居和Tunnel

```
<R1>display  interface Tunnel
Tunnel0
Current state: UP
Line protocol state: UP
Description: Tunnel0 Interface
Bandwidth: 64 kbps
Maximum transmission unit: 1476
Internet address: 192.168.3.1/24 (Primary)
Tunnel source 100.1.1.1 (GigabitEthernet0/0), destination 200.2.2.3
Tunnel keepalive disabled
Tunnel TTL 255
Tunnel protocol/transport GRE/IP
    GRE key disabled
    Checksumming of GRE packets disabled
Output queue - Urgent queuing: Size/Length/Discards 0/1024/0
Output queue - Protocol queuing: Size/Length/Discards 0/500/0
Output queue - FIFO queuing: Size/Length/Discards 0/75/0
Last clearing of counters: Never
Last 300 seconds input rate: 0 bytes/sec, 0 bits/sec, 0 packets/sec
Last 300 seconds output rate: 0 bytes/sec, 0 bits/sec, 0 packets/sec
Input: 44 packets, 3252 bytes, 0 drops
Output: 45 packets, 3308 bytes, 0 drops

<R1>display  interface Tunnel brief
Brief information on interfaces in route mode:
Link: ADM - administratively down; Stby - standby
Protocol: (s) - spoofing
Interface            Link Protocol Primary IP      Description
Tun0                 UP   UP       192.168.3.1
```

```
<R1>display  ospf peer

         OSPF Process 1 with Router ID 192.168.3.1
               Neighbor Brief Information

 Area: 0.0.0.0
 Router ID       Address         Pri Dead-Time  State             Interface
 200.2.2.3       192.168.3.3     1   35         Full/ -           Tun0
 ----
 <R3>display  interface Tunnel
Tunnel0
Current state: UP
Line protocol state: UP
Description: Tunnel0 Interface
Bandwidth: 64 kbps
Maximum transmission unit: 1476
Internet address: 192.168.3.3/24 (Primary)
Tunnel source 200.2.2.3 (GigabitEthernet0/0), destination 100.1.1.1
Tunnel keepalive disabled
Tunnel TTL 255
Tunnel protocol/transport GRE/IP
    GRE key disabled
    Checksumming of GRE packets disabled
Output queue - Urgent queuing: Size/Length/Discards 0/1024/0
Output queue - Protocol queuing: Size/Length/Discards 0/500/0
Output queue - FIFO queuing: Size/Length/Discards 0/75/0
Last clearing of counters: Never
Last 300 seconds input rate: 0 bytes/sec, 0 bits/sec, 0 packets/sec
Last 300 seconds output rate: 0 bytes/sec, 0 bits/sec, 0 packets/sec
Input: 51 packets, 3720 bytes, 0 drops
Output: 51 packets, 3728 bytes, 0 drops

<R3>display  interface Tunnel brief
Brief information on interfaces in route mode:
Link: ADM - administratively down; Stby - standby
Protocol: (s) - spoofing
Interface            Link Protocol Primary IP      Description
Tun0                 UP   UP       192.168.3.3

<R3>display  ospf peer

         OSPF Process 1 with Router ID 200.2.2.3
               Neighbor Brief Information

 Area: 0.0.0.0
 Router ID       Address         Pri Dead-Time  State             Interface
 192.168.3.1     192.168.3.1     1   35         Full/ -           Tun0
<R3>
```

查看路由表

```
<R1>display  ip routing-table
```

```
Destinations : 18       Routes : 18

Destination/Mask    Proto   Pre Cost        NextHop         Interface
0.0.0.0/0           Static  60  0           100.1.1.2       GE0/0
0.0.0.0/32          Direct  0   0           127.0.0.1       InLoop0
100.1.1.0/24        Direct  0   0           100.1.1.1       GE0/0
100.1.1.1/32        Direct  0   0           127.0.0.1       InLoop0
100.1.1.255/32      Direct  0   0           100.1.1.1       GE0/0
127.0.0.0/8         Direct  0   0           127.0.0.1       InLoop0
127.0.0.1/32        Direct  0   0           127.0.0.1       InLoop0
127.255.255.255/32 Direct  0   0           127.0.0.1       InLoop0
192.168.1.0/24      Direct  0   0           192.168.1.1     GE0/1
192.168.1.1/32      Direct  0   0           127.0.0.1       InLoop0
192.168.1.255/32    Direct  0   0           192.168.1.1     GE0/1
192.168.2.0/24      O_INTRA 10  1563        192.168.3.3     Tun0
192.168.3.0/24      Direct  0   0           192.168.3.1     Tun0
192.168.3.1/32      Direct  0   0           127.0.0.1       InLoop0
192.168.3.255/32    Direct  0   0           192.168.3.1     Tun0
224.0.0.0/4         Direct  0   0           0.0.0.0         NULL0
224.0.0.0/24        Direct  0   0           0.0.0.0         NULL0
255.255.255.255/32 Direct  0   0           127.0.0.1       InLoop0
---
<R2>display  ip routing-table

Destinations : 15       Routes : 15

Destination/Mask    Proto   Pre Cost        NextHop         Interface
0.0.0.0/32          Direct  0   0           127.0.0.1       InLoop0
100.1.1.0/24        Direct  0   0           100.1.1.2       GE0/0
100.1.1.2/32        Direct  0   0           127.0.0.1       InLoop0
100.1.1.255/32      Direct  0   0           100.1.1.2       GE0/0
127.0.0.0/8         Direct  0   0           127.0.0.1       InLoop0
127.0.0.1/32        Direct  0   0           127.0.0.1       InLoop0
127.255.255.255/32 Direct  0   0           127.0.0.1       InLoop0
192.168.1.0/24      Static  60  0           100.1.1.1       GE0/0
192.168.2.0/24      Static  60  0           200.2.2.3       GE0/1
200.2.2.0/24        Direct  0   0           200.2.2.2       GE0/1
200.2.2.2/32        Direct  0   0           127.0.0.1       InLoop0
200.2.2.255/32      Direct  0   0           200.2.2.2       GE0/1
224.0.0.0/4         Direct  0   0           0.0.0.0         NULL0
224.0.0.0/24        Direct  0   0           0.0.0.0         NULL0
255.255.255.255/32 Direct  0   0           127.0.0.1       InLoop0
---
<R3>display  ip routing-table

Destinations : 18       Routes : 18

Destination/Mask    Proto   Pre Cost        NextHop         Interface
0.0.0.0/0           Static  60  0           200.2.2.2       GE0/0
0.0.0.0/32          Direct  0   0           127.0.0.1       InLoop0
127.0.0.0/8         Direct  0   0           127.0.0.1       InLoop0

127.0.0.1/32        Direct  0   0           127.0.0.1       InLoop0
```

```
127.255.255.255/32 Direct  0   0           127.0.0.1       InLoop0
192.168.1.0/24     O_INTRA 10  1563        192.168.3.1     Tun0
192.168.2.0/24     Direct  0   0           192.168.2.1     GE0/1
192.168.2.1/32     Direct  0   0           127.0.0.1       InLoop0
192.168.2.255/32   Direct  0   0           192.168.2.1     GE0/1
192.168.3.0/24     Direct  0   0           192.168.3.3     Tun0
192.168.3.3/32     Direct  0   0           127.0.0.1       InLoop0
192.168.3.255/32   Direct  0   0           192.168.3.3     Tun0
200.2.2.0/24       Direct  0   0           200.2.2.3       GE0/0
200.2.2.3/32       Direct  0   0           127.0.0.1       InLoop0
200.2.2.255/32     Direct  0   0           200.2.2.3       GE0/0
224.0.0.0/4        Direct  0   0           0.0.0.0         NULL0
224.0.0.0/24       Direct  0   0           0.0.0.0         NULL0
255.255.255.255/32 Direct  0   0           127.0.0.1       InLoop0
<R3>
```